ERDENES POWER LLC

Erdenes power LLC is formed solely for the purpose developing this project and currently owned 100% by two individuals who have power industry and financial backgrounds and expertise respectively. The project development is funded by the involved individuals and there is no any current liabilities at the company level.

The Project Location

Jargalant Valley, Tsenkhermandal Soum, Khentii Province

Location of the Project Site

The project location is in area of Khentii mountain systems at the very beginning of Kherlen river which has total length over 1,200 km before it discharges into lake Dalai in China. Initially, the lower reservoir of the project facility stores the necessary water for the operation of the facility that is less than 1% of the annual discharge of the river. The project site is in 150 km from Ulaanbaatar city and to be interconnected with the Baganuur 220/110/10 kV substation which is located in 15 km.

The land necessary for the project is acquired and secured from the local authorities and it does not overlap with any private properties, mining licenses and critical infrastructure. No any people's reallocation is required. Baganuur, the coal mining town which is administrative district of Ulaanbaatar city, is interconnected to Ulaanbaatar through paved road, railway and 220kV tx lines.

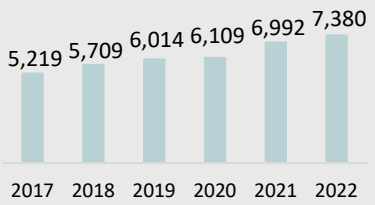
Rationale of the project

The Central Power Grid, backbone of the Mongolian power energy sector generates 76% of the country's total electricity consumption, distributes and supplies power to meet 95% of the power demand of the country and its power generation capacity consists of mainly base-load type coal-fired combined heat and power generation power plants that are aged over 40 years. The only flexibility in dispatching of power and regime regulation comes from the power import from Russian grid through 250 km double-circuit 220 kV transmission lines. These lines have been loaded to maximum capacity in recent years, with the current import capacity contract at 350 MW.

Although, there were significant additions to the generation capacities of the Central Grid through renewables i.e. solar and wind farms in the last decade; although, they do not contribute to the extent the needed to the grid as far as the regime regulation and reliable dispatchable power capacity. The present project, 45MW pumped storage project facility offers to the grid reliable dispatchable power that has been much sought for to meet the peak load demand of the grid. Whilst substituting the import of power, the project facility utilizes and stores the off-peak excess power generation of the coal-fired power plants that are currently being exported to the Russian grid at the fraction of the cost of import thereby it creates overall efficiency to the grid operation apart from the obvious economic benefits to the Central Grid. Being geographically located at the south-east conjunction of the Central Grid, the project facility contributes in reducing technical losses along the transmission lines extended to 500-600 kms toward south and east and adding grid safety and reliability in these areas to certain extent.

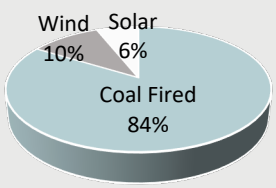
There have been a number of attempts from the Mongolian Government to construct classic dam type hydro projects on the rivers with more of hydro power potentials in the country that are exclusively outflowing to Russian territory. However, these projects were blocked by Russian Government at the bilateral and intergovernmental implications to the lake Baikal's eco-systems.

Power Demand in Central Grid of Mongolian Energy System in mn.kWh



Installed Capacity

Coal Fired	1,233MW
Wind	150MW
Solar	90MW
Total	1,478MW



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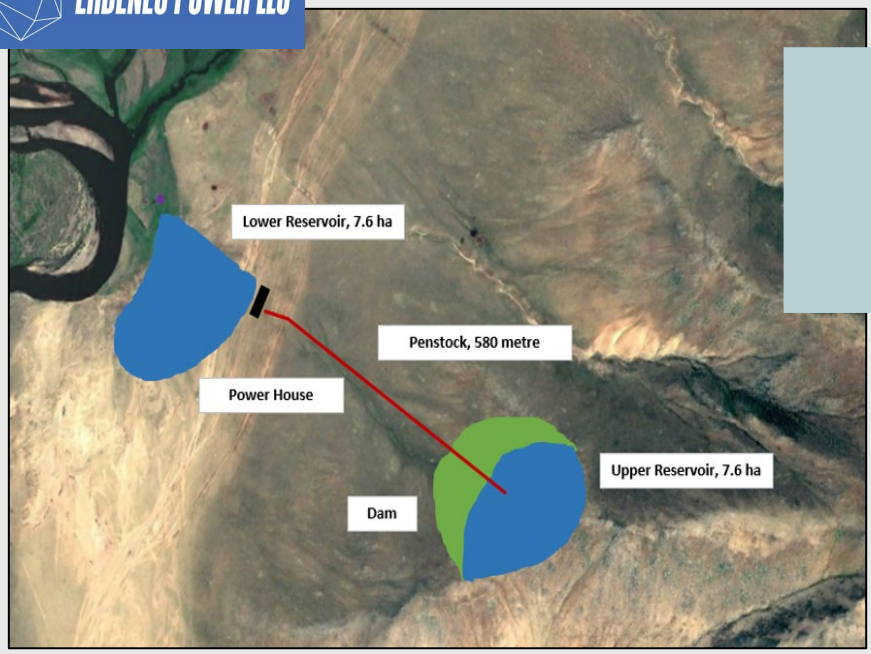
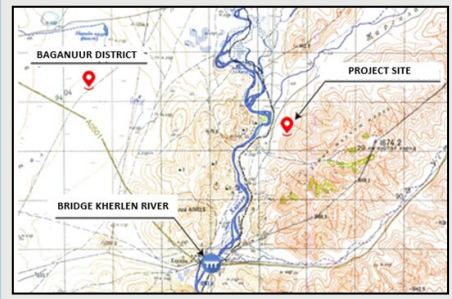


The Pumped Storage Facility

Technical Parameters

Gross head:	138 m
Water volume:	1.8mn cubic m
Charge hours:	6.75
Discharge hours:	8 hours
Penstock:	580 m

The pumped storage plant consists of facilities and major equipment including lower and upper water reservoirs, water intake, penstocks, 3 units of 15MW vertical Francis type turbine-generators reversible in pump mode with more than 80 percent efficiency with other auxiliaries like switch yard, distributed control system, in-house power distribution, etc. General layout and/or arrangement of equipment and preliminary design of the facility are made available. The design of 110kV double circuit overhead transmission lines (OHL) for interconnection to the grid is made and has got expertise approval from the relevant government authorities.



152 km east of Ulaanbaatar,
12 km east of Baganuur Town,
6.7 km north of Kherlen River Bridge

Permits and licenses of the project

Power storage and generation facility construction license and power purchase and supply tariffs are granted and approved by the Energy Regulatory Authority (ERC) back in 2021 based on feasibility study approval and the Ministry of Energy authorization. Amounts of energy and respective tariffs for the power purchase from the Central Grid for pumping up is 138.54 mn.kWh per annum at 0.0252 US$/kWh and for supply to the Central Grid is 110.83 mn.kWh per annum at 0.101 US$/kWh. Subsequently, the Power Purchase and Supply Agreement (PPA) is negotiated and signed with the National Dispatch Center, SOE for 15 years from the commercial operation date at the approved tariffs in 2022.

The necessary land of the project operation and construction of the OHL routing are acquired and secured.

The Detailed Environmental Impact Assessment is developed and approved by the Ministry of Environment and Tourism. River water usage and construction of hydro power facilities at the river bay within 150 meters are approved and authorized by the local parliament, river protection administration, local community an central government branches.

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Financial indicators

The financial model of the project returns IRR at 12.6% and NPV of $8.6M USD at the 8% discount rate with the assumptions of debt leverage at 70% with financing 6% interest rate per annum for 15 years including 2 years of grace period with estimate $50M USD capital expenditure.

IRR	NPV	CAPEX
12.6%	$8.6M	$50M

Project Capital Expenditure Estimation

Construction	
Lower Reservoir	10.5M USD
Upper Reservoir	3.8M USD
Penstock	11.7M USD
Plant	5.5M USD
Water Intake Facility	0.1M USD
Pump Turbine	14.5M USD
OHL	2.2M USD
Substation and other	1.7M USD
Total	**50.0M USD**

Project Opportunities

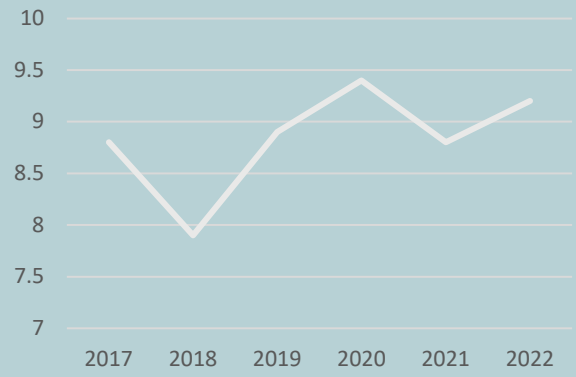
The Government of Mongolia supports and promotes foreign investment in the power sector as the industry struggles to meet the growing power demand at the rate of **7.2%** last five years. The custom taxes for the import renewal energy equipment has been eliminated and income tax for the renewable energy generation has been excluded from income tax for first 3 years of operation and reduced to **50%** for the subsequent 3 years.

Russian import tariff is continuously increasing in the last three years and **10%** increase in capacity charges and **30%** in energy tariffs of import from Apr 1, 2023 is imminent; moreover, the import transmission line has reached its technical maximum.

Average Import Tariffs of the Central Energy System on Annual basis

Average Import Tariffs in US$/kWh

Winter period from 2021 to 2022: 10.61 cents
Winter period from 2022 to 2023: 10.73 cents



Lastly, the project geographical location allows to increase the capacity of the storage in the future at least doubling the capacity by adding the same equipment which is assumed to be expedited processing from the permitting perspective that also will increase efficiency of overall investment.

The government is in continuous search of energy independence including power apart from fossil fuels such as gasoline, diesel, heating oil, natural gas, etc., which are major import items in trade balance of the country.

The government is open to make amendments and revisions to the project PPA that deem necessary to the investor; however, it requires longer timespan as the protocol/procedure foresees various government instances involved in.